Attribution Framing by Outlet Political Lean: A Companion Analysis

Hariaksha Gunda

August 23, 2026

Purpose

This memo supplements Section 5.5 ("Variation by Outlet Characteristics") of the main paper, *Reading the Storm*. That section reports a regression of article-level attribution scores on a continuous outlet political-lean measure (Ad Fontes Bias score). This memo presents the same relationship visually, as a time trend split by outlet political lean, and documents why one of the three lean groups — Right-leaning outlets — is represented by only a small number of articles.

This is a working memo, not a paper section: it is meant to inform a conversation with the coauthor team about how (or whether) to extend the outlet-lean data collection, not to be read as a finished result.

Methodology

Outlet political lean is determined exactly as in the main paper. Each article's URL is parsed down to its bare domain (e.g. `nytimes.com`), which is then looked up against Ad Fontes Media Bias Chart's numeric "Bias" score — a continuous scale from roughly −42 (furthest left) to +42 (furthest right) — for the 33 domains in the sample that Ad Fontes has individually rated. An article is classified as:

- **Left-leaning** if its domain has a negative Bias score,
- **Right-leaning** if its domain has a positive Bias score,
- **Unidentified** if its domain has no individually-rated Bias score (either because the domain was never rated, or because it was rated but no numeric score was available

— one domain, `al.com`, falls in this category due to a data error on Ad Fontes' page for that outlet).

These lean labels, along with the underlying numeric Bias score, are now stored directly as columns on the `Master` sheet of `attribution.xlsx` (`Domain`, `Ad Fontes Bias Score`, `AllSides Rating (Numeric)`, `Political Lean Group`), rather than being computed only at analysis time.

Unlike Figure 1 of the main paper, which plots the disaster-level Attribution Index (one point per disaster), this analysis plots the *article*-level attribution score $a_{i,d}$, since outlet lean only varies at the article level — the same disaster is frequently covered by both a left- and a right-leaning outlet, so collapsing to one number per disaster would discard the variation this chart is meant to show.

Sample composition

Table 1 shows how the 1,252 Relevant articles split across the three lean groups.

Group	Articles	Share
Left-leaning	869	69.4%
Unidentified	363	29.0%
Right-leaning	20	1.6%

Table 1: Relevant articles by outlet political-lean group.

The Right-leaning group is thin: only 6 of the 33 individually-rated domains in the sample have a positive Ad Fontes Bias score (Deseret News, New York Post, OC Register, Boston Herald, Wall Street Journal, Las Vegas Review-Journal), and together they contribute just 20 articles across the full 2000–2024 sample — roughly half of them from Deseret News alone.

Chart

Figure 1 plots the annual article-level mean attribution score for Left-leaning and Unidentified outlets (± 1 SE, with the article count n labeled at each point), and a binned mean for Right-leaning outlets. Because Right-leaning coverage is so sparse — several individual years have zero Right-leaning articles at all — an annual mean for that group would mostly be undefined or based on 1–2 articles. Instead, Right-leaning articles are pooled into three periods (2000–2019, 2020–2021, 2022–2024), chosen only to keep each bin's article count usable ($n = 5$, $n = 9$, $n = 6$ respectively), not because of any hypothesized break at those specific years.

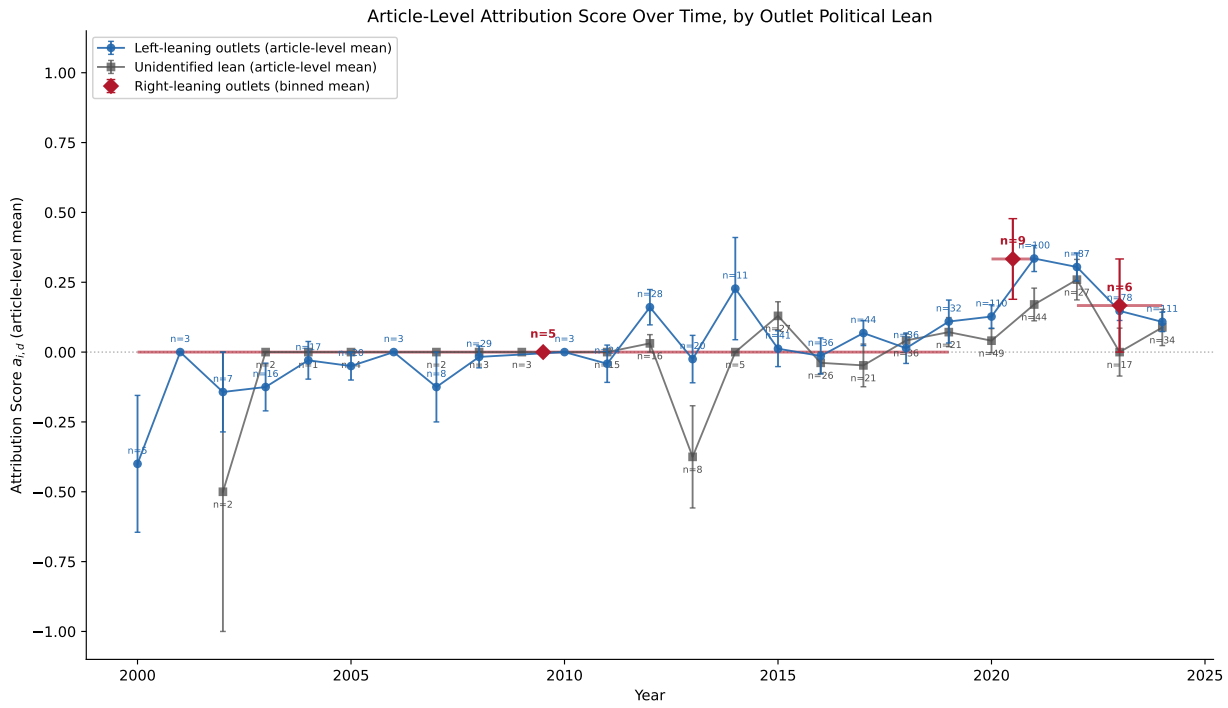


Figure 1: Article-level attribution score over time, by outlet political lean. Left-leaning and Unidentified are annual means (±1 SE); Right-leaning is a binned mean across three periods given its small sample.

Reading this chart

Left-leaning and Unidentified outlets show broadly the same pattern documented in the main paper's Figure 1: near-zero or negative average attribution scores through the mid-2000s, rising toward modestly positive values after 2010, with a further increase from roughly 2020 onward. The Right-leaning group's binned mean is directionally similar — near zero pre-2020, rising in the two most recent bins — but this should be treated as illustrative rather than statistically reliable, given the sample sizes involved. The widest error bar in the figure (the 2020–2021 Right-leaning bin) still rests on only 9 articles.

Why is Right-leaning representation so thin?

We investigated whether this imbalance reflects a fixable issue in the article-retrieval pipeline (`code/google_search.py` and `code/build_state_engine_mapping.py`) rather than the LLM coding step. Three findings:

1. **It is not a coding-stage bias.** Candidate articles from Right-leaning domains convert to `Relevant` at a 100% rate (every candidate article retrieved from a Right-leaning

domain was coded Relevant), actually higher than the 60–92% conversion rate for the largest Left-leaning domains. The LLM relevance-screening step is not filtering out Right-leaning content disproportionately — the shortfall happens upstream, at retrieval.

2. **Fox News, despite being in the search domain list, returned zero candidate articles across all 313 events and 24 years.** Fox News (`foxnews.com`) and the Wall Street Journal (`wsj.com`) are both explicitly included in the "National" Programmable Search Engine's domain list, and that engine is queried for every single disaster event (confirmed in `build_state_engine_mapping.py`: the National engine is added to every event's engine set unconditionally). Yet `foxnews.com` produced 0 candidate articles, and `wsj.com` produced only 1, across the entire sample. The likely mechanism: the Google Custom Search API returns only the top 10 organic results per query, and a single Programmable Search Engine spans dozens of domains at once with no live per-domain restriction — so for a given disaster-name query, Google's own ranking decides which domains fill those 10 slots. Wire-syndicated coverage that gets republished verbatim across the *New York Times*, *Washington Post*, CNN, NPR, *USA Today*, and AP tends to dominate those slots for exact-phrase disaster-name queries, crowding out Fox News' and WSJ's own (differently worded) coverage of the same events, even though both are eligible domains that get queried every time.

3. **The domain list itself under-represents Right-leaning outlets.** As confirmed separately, the Access World News database's own "national" tagging covered fewer than 10 genuinely national outlets; the remainder of the 47-domain National PSE list appears to have been swept in by a fallback rule (domains with no populated "State" field defaulting to the National bucket) rather than deliberate curation. Well-known Right-leaning national outlets that were never AWN-tagged at all — Breitbart, the Washington Examiner, National Review, the Daily Wire, Newsmax, the Daily Caller — are absent from the domain list entirely, so they were never eligible to be retrieved in the first place. No script currently in `code/` generates `data/pse_domains.json`; it appears to be a static artifact from earlier in the project rather than a reproducible output.

Possible next step

If the coauthor team wants better Right-leaning representation, the most direct fix is a dedicated Programmable Search Engine restricted to a curated list of Right-leaning national

outlets (e.g. Fox News, WSJ, New York Post, Washington Examiner, National Review, the Daily Wire, Newsmax, the Daily Caller, Breitbart, Washington Times), so those outlets compete against each other for search-result slots rather than against *The New York Times* and *Washington Post*. This would require:

- Creating a new Google Programmable Search Engine (a manual step in Google's console — not something that can be done from the existing codebase) restricted to the agreed domain list;
- Re-running the existing collection pipeline (`google_search.py`) against just that engine across all 313 events;
- Coding the resulting candidates through the existing LLM pipeline (`llm_coder.py`) for consistency with the rest of the corpus;
- Deciding, as a team, how to treat the addition methodologically — since it would mean the Right-leaning subsample was collected differently (targeted rather than incidental) from the rest of the corpus, which is worth flagging explicitly if it ends up in the paper.

This memo does not take a position on whether that extension is worth doing; it is meant to give the team the diagnostic information needed to decide.